

Energy Coaching in Massachusetts - Final GIS Report

Introduction

Context & Background

Energy coaching in Massachusetts is a growing practice where municipalities and nonprofits hire coaches to help residents understand their home heating and cooling systems and where they can be improved. This usually means talking through heat pumps as an alternative to older systems and explaining the benefits, such as lower costs, better comfort, and reduced environmental impact. These conversations are meant to be low pressure and informational rather than a sales pitch. Coaches rely on both people skills and technical knowledge in order to be effective.

Across Massachusetts, the availability of energy coaching varies from place to place. Some communities have established coaching programs, while others only have limited support, short-term support, or no support at all. This uneven access reflects broader disparities between wealthier and poorer communities. Residents in lower income municipalities often deal with older systems and higher energy burdens but have fewer opportunities to receive guidance on upgrades and resources available to them. These differences are part of why I am focusing on energy coaching, because of its role in shaping who receives information, support, and resources during the transition to cleaner heating and cooling that we see accelerating as people grow concerned about the planet, their bills, and their comfort.

Problem Statement

I was curious about exploring energy coaching in Massachusetts more deeply because I've learned through my internship with Northeast Energy Efficiency Partnerships (NEEP) that not all municipalities in MA have energy coaching available, and from the ones that I knew do have coaching, they are known to be more affluent. So when thinking about it from an equity

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perspective, which has been central to NEEPs recent mission, I thought about tangibly pulling together data from NEEP with publicly available population, income and municipal boundary data. This would allow me to more directly start exploring where there could be equity gaps in the Massachusetts energy coaching landscape as it currently exists if low & moderate income populations can't easily access the resources that are meant for them to adopt heat pumps. From this line of thinking, my **research question** emerged, which is:

- **How do the median incomes (per capita) of municipalities with energy coaching programs differ from those without coaching programs?**
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Methodology

How did I analyze this problem?

As mentioned previously, I used publicly available data, specifically from the official Massachusetts websites. The median income per capita & population data was from Massachusetts Department of Revenue (MassDOR), the municipal boundary data was from MassGIS, and the coaching data was from NEEP, my internship host and a regional non-profit helping advance energy policy in the Northeast.

Starting with MassDOR, I downloaded the data and opened it in Excel, removing all of the columns that were not needed for my analysis, such as EQV and EQV per capita. Next, I downloaded the data from NEEP and started cleaning it. The sheet had listed what municipalities had coaching, but it was not in a way that was analyzable. So I had to change the data to have a column called "Coaching?" that was a simple Y or N if it did or did not have coaching. Then, I copied this over into the MassDOR sheet and saved it. Lastly, I opened the municipal boundary data and the tables in GIS, performed a join, and had the foundation for the rest of my analysis.

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Scale & Accuracy of Data

For each of my datasets, there were 351 municipalities to analyze. This makes it sufficiently large enough for analysis without being too small to derive meaningful results or too large to be able to properly analyze at all. Thankfully, my data seems pretty accurate as well, given that 2 of 3 are from official Massachusetts sources. My data source on coaching is also pretty accurate, though I do have some thoughts about how it could be better/more accurate. These concerns arise from the fact that it is from August 2025, and I know that more municipalities are in the process of adopting energy coaching programs. Therefore, my data could already be slightly outdated. But, it is from recent enough that this is not of enough concern to cause me to disregard the data altogether.

Limitations of the Data

One of the major limitations of my coaching data from NEEP is that it did not easily distinguish between the coaching programs that are administered by the town and those that are administered by a non-profit. The reason that this is important to consider is because I've learned through my internship that places that use municipal coaching seem to have greater success and resident buy-in than places that use coaching administered by a non-profit. Having this data would have allowed for another interesting layer of analysis, comparing the income of municipalities that use their own coaches versus those that use external coaches to see if income could be having more of an effect on buy-in rather than the coaching type.

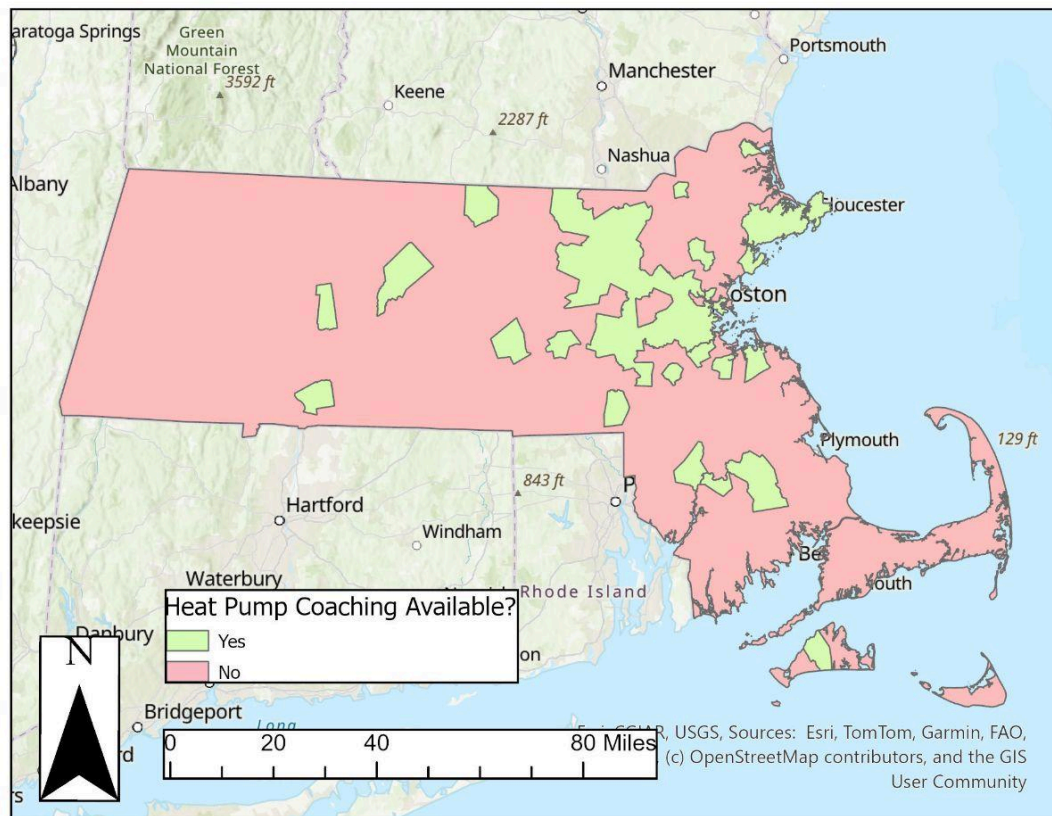
Regardless of this critique though, all of my data together has few limitations and allows for some pretty interesting analysis.

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Analysis

Below you'll find a map that is useful for some background information before getting into the substantial analysis. It is meant to provide clarity on where coaching already is and how many people do or do not have access to it.

Areas with & without Energy Coaching in MA



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Population w/ Coaching	Population w/ out Coaching
2,773,015 people *675,647 people in Boston alone	4,256,902 people
39.45% of MA population	60.55% of MA population
62 municipalities	289 municipalities

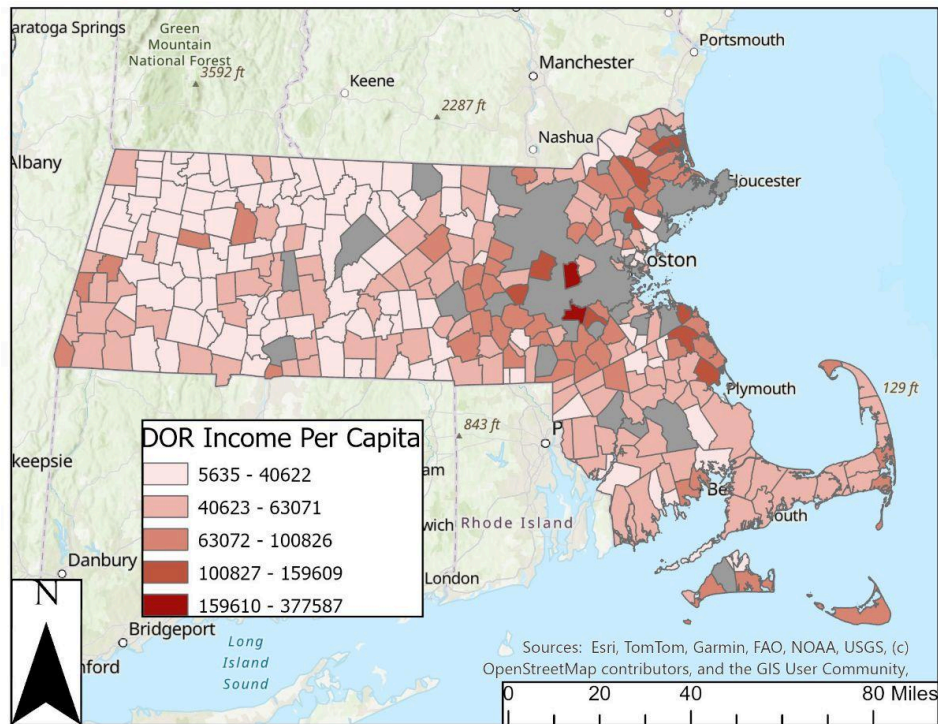
I got these numbers by performing the summary statistics function on ArcGIS, which allowed me to get the sum of the different columns in the table. I also used the dissolve function to remove all of the municipal boundaries to make the map a little clearer.

This initial analysis was useful for getting a sense of how many people already have energy coaching coverage where they live versus those who do not. Because energy coaching is implemented at the municipal level, it can lead to the patchwork of coverage that we see above. This then allows us to follow up this analysis with analysis of each area independently to understand their differences and where policies should or should not be targeted in order to be most effective.

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Analysis of municipalities without coaching

Munis without Energy Coaching in MA & Median Incomes per capita



Distribution of No Coaching Incomes

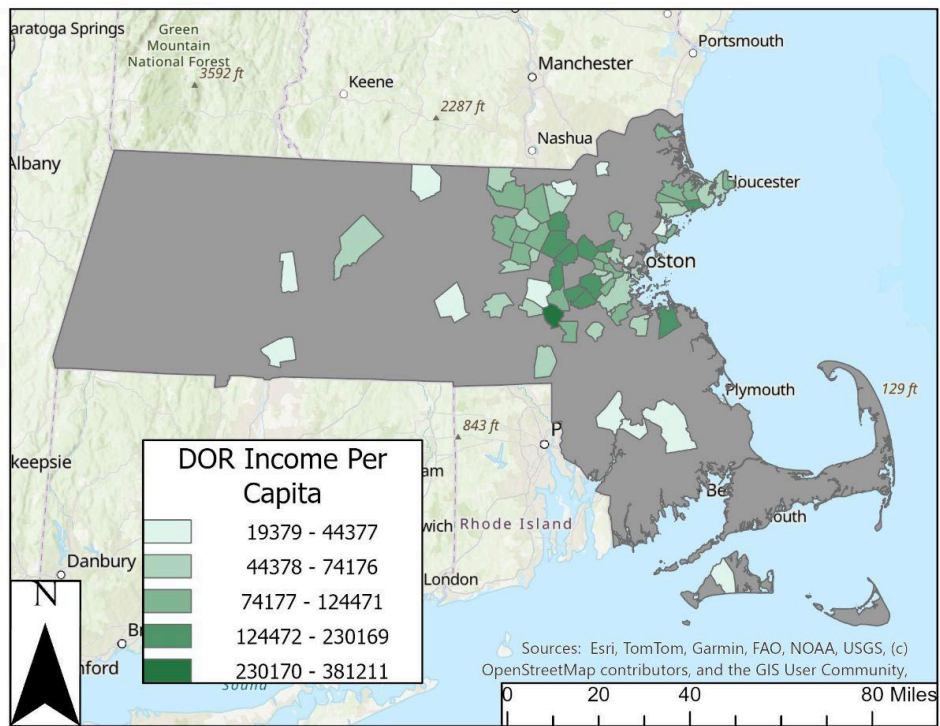


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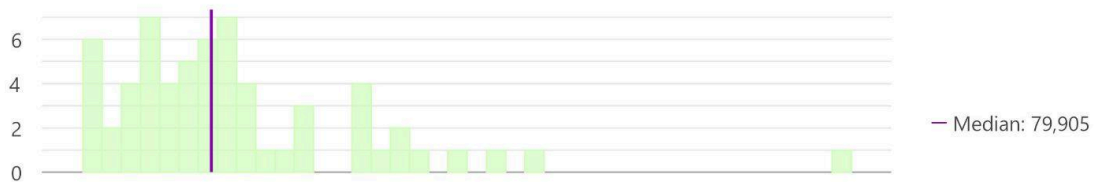
The median value on the chart above (45,421) was calculated using the histogram tool on the 289 municipalities that do not have coaching available. On its own, this map does not tell us much. However, when we compare it to the municipalities that do have coaching, along with statewide median income, we start to see a much more interesting picture.

Analysis of municipalities with coaching

Munis with Energy Coaching in MA & Median Incomes per capita



Distribution of Yes Coaching Incomes



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The median value on the chart above (79,905) was calculated using the histogram tool on the 62 municipalities that have coaching available. When compared to the map and histogram of the municipalities without coaching, an incredibly interesting picture emerges.

Results

The analysis confirms the musings I'd heard at my internship; that municipalities with coaching are indeed wealthier than the municipalities without. Many of the munis with coaching are grouped around the metrowest region and the coastal North Shore, which are both areas of Massachusetts that are known to be wealthier. Looking specifically at the numbers reveals just how stark the differences are between munis with and without coaching.

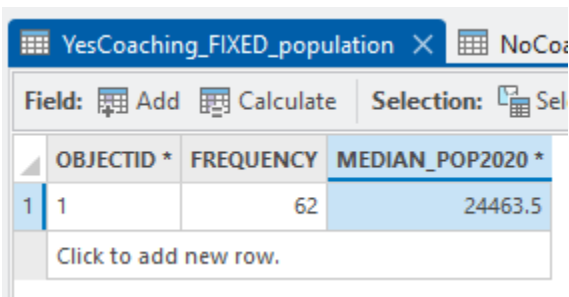
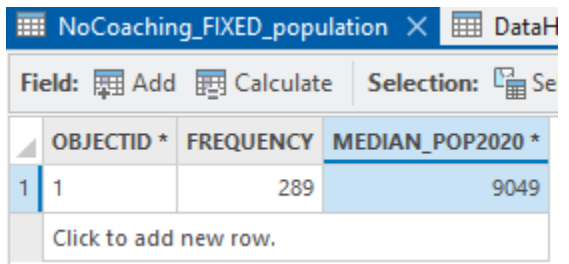
As shown in my analysis, municipalities without coaching have a median income per capita of 45,421. This is starkly different from my results for munis with coaching, which had a median income per capita of 79,905, or over 34,000 dollars higher than munis without coaching. In terms of what this means, it means quite a lot. It means that access to energy coaching in Massachusetts is concentrated in more affluent areas that, while it's good that they have coaching, are not serving the populations who have the highest need for access to resources and programs that help them save money and live more comfortably.

Another layer of analysis I decided to do briefly was to calculate the median income per capita for all of Massachusetts, and compare it to these values. For no coaching, again the value was 45,421. For yes coaching it was 79,905. For all of Massachusetts, it was 47,294. What this means is that munis without coaching are only slightly below the median for all of MA, while munis with coaching are far above it. This supports my conclusion that coaching is concentrated in wealthier municipalities.

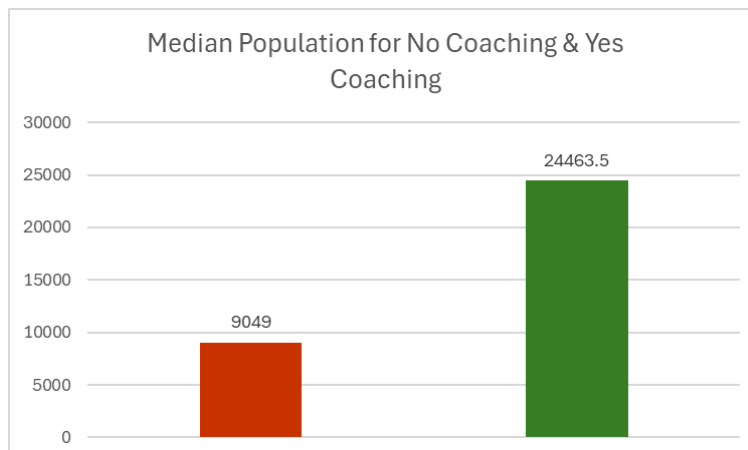
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As can be seen, Eastern MA is the population hotspot. This was already known, but this analysis confirms it. And, as mentioned earlier, most of the coaching towns in MA were towards the east, so I'm interested in seeing what the median populations are like since it's now clear that many are close to, or within, the orange/red on the map.

Using the summary statistics tool once again, I got a median value for no coaching munis of 9,049 people. For yes coaching munis, I got a median value of 24,463.5 people, or over 2.5x higher than the median population for no coaching municipalities.

Yes Coaching Output Table	No Coaching Output Table
 The screenshot shows a table titled 'YesCoaching_FIXED_population'. It has three columns: 'OBJECTID *', 'FREQUENCY', and 'MEDIAN_POP2020 *'. The first row shows an OBJECTID of 1, a frequency of 62, and a median population of 24463.5. Below the table is a button that says 'Click to add new row.'	 The screenshot shows a table titled 'NoCoaching_FIXED_population'. It has three columns: 'OBJECTID *', 'FREQUENCY', and 'MEDIAN_POP2020 *'. The first row shows an OBJECTID of 1, a frequency of 289, and a median population of 9049. Below the table is a button that says 'Click to add new row.'

Here is a bar chart that I made in Excel showing the scale of how different these populations are



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This is a pretty interesting and relevant finding because it shows that the population of the municipalities with coaching are generally substantially higher than the municipalities without. Combine this with the findings on income being substantially higher as well, you get a combination that shows coaching municipalities are more affluent and more populated. This makes sense given something else I previously mentioned, which is how coaching towns are largely clustered around the metrowest region and North shore, which are well known to be affluent suburbs in Eastern Massachusetts.

In the end, I was able to show that the median incomes per capita for municipalities with existing coaching programs are much higher than those without coaching programs. And in addition to this, I was able to explore a deeper layer of understanding as to why coaching programs might have been implemented, that being population. Larger municipalities have more resources, and so it would make sense that they are the first to adopt such a program. But for the places that haven't, there are still lessons to be learned and ways to adapt existing programs into potential models for these other areas.

Conclusion

Policy Recommendations

From my analysis, my main recommendation would be for Massachusetts to do another grant program similar to the Municipal Energy Manager (MEM) Grant from earlier this year. The MEM Grant is a program that is funding 3-year energy manager positions for municipalities and regional planning commissions to increase energy efficiency program and policy adoption, particularly in schools. (The RCS Network, n.d.). The policies should focus on providing

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municipalities with the resources and toolkits to implement a coaching program, ideally with some funding attached as well. Maybe building on the MEM grant, these energy managers can implement a coaching program in their municipality or region if it does not have one already.

My findings on population, and the structure of the MEM grant, can provide more clear policy guidance as well. The MEM grant provides 24 energy managers for 63 municipalities, which you'll quickly notice is not a 1:1 ratio. This is because some of the energy managers cover multiple municipalities through the regional planning commissions that they belong to, such as the Hampshire Regional Partnership in Central/Western MA. This could be a model that energy coaching uses as well, collecting many of the smaller, less affluent towns into these groups that are the coverage area for the potential coaches. This also helps group demand together, because having a coaching program for municipalities with fewer than 10,000 people would likely, by my best guess, not generate enough demand to rationalize having the program. However, having 4, 5, 6, or more municipalities grouped together would create a population that rivals similarly sized municipalities with proven coaching programs. And while the income of these municipalities are shown to be lower than that of eastern MA, MassSave has different incentive structures that are meant to be accessible to lower income communities as well. Therefore, it's really coaching implementation that would act as a bridge between these areas and resources they would likely benefit from.

These policies need to not only give municipalities the resources to implement a coaching program, but also provide guidance on how to recruit and retain coaches. Many coaching programs in their current state rely on volunteer coaches. This can be good because it's cheaper, but volunteers tend to be less reliable and more prone to burnout if they are overworked. Therefore, the policies should provide guidance on balancing the realities of most coaches being

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volunteers (such as having smaller time commitments), while ensuring that work is still given to them so they maintain consistent engagement as a coach. Too much work and they'll likely leave, same with too little. Maybe municipalities attempting to implement coaching can reach out to similarly sized municipalities with coaching programs and ask for their wisdom. As seen through my internship where I met to discuss coaching with municipal staff from Acton, Concord, Wellesley, Marblehead, Winchester, Needham, etc, this is a process that can greatly benefit from collaboration and discussion. If municipalities do not have the resources to implement the coaching directly, there are also organizations like HeatSmart Alliance that can help fill in the gap with coaching (HeatSmart Alliance, 2025), though again, even if non-profits can be and often are successful, there's anecdotal evidence that directly administered municipal programs have better outcomes.

Next Steps

Thinking about what more could be done, I would like to spend more time on the difference between programs administered by non-profits versus municipalities to see if there are any meaningful differences that could be explained by the data. From that research, I'd be able to better explore what the successes and limitations of each type are and how they can be strengthened overall to better meet the needs of Massachusetts.

Something else I have also learned from my internship this semester is that Massachusetts, even though its coaching programs have problems, is still a pioneer in this space nationally. Very few other states have programs as widespread or robust as those that exist here in Massachusetts, so that's worth noting. This leads me to another avenue of research that I could pursue, which is comparing Massachusetts programs to those of other states, especially in the NEEP region (Delaware, Washington DC, Connecticut, Maine, Maryland, Massachusetts, New

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Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, Vermont, & West Virginia). I have researched energy coaching in my internship briefly for Connecticut, and what I found was that the programs that exist there are also fragmented but administered by a more corporate entity rather than having someone “local” give the consultation.

What to do differently

As far as what I’d change or do differently, I’d say I wish I found more time to actually visualize the differences in median incomes per capita onto my map more clearly. While I am satisfied with the maps that I made, I feel that making them integrate the medians more effectively would have made the maps more impactful overall. I’d also do more to explore the non-profit versus municipal coaching program administration, as well as asking my internship manager if there were more data on coaching program outcomes to see if I could map “success” of coaching programs in some way in order to find a possible common thread that connects them all, potentially leading to clearer policy ideas for other places. However, I’m happy with the hotspot analysis I did for the population, as it was able to add another layer of analysis that I think was useful for understanding.

Regardless and more generally, I think I’d also like to try and find more projects that use raster data rather than only vector data. However, I’m having trouble thinking about projects that connect my professional interests with a project that would actually use raster data.

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Coaching data is from my internship supervisor, Andy Winslow, and not an official source, though he and I work for Northeast Energy Efficiency Partnerships and that is the avenue through which the data was shared.